Sajad Abavisani

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SUMMARY

Ph.D. in Electrical & Computer Engineering (Georgia Tech, 2026) specializing in causal inference and statistical modeling on noisy multichannel time series: telling true causes apart from correlated symptoms when data are sampled more slowly than the process that generates them. First-author papers at NeurIPS 2026 and ICLR 2023 (notable top 25%, oral). Builds end-to-end systems, including exact combinatorial solvers that scale to 100+ variables, transformers trained on ~1M simulated series, and LLM fine-tuning and evaluation pipelines, and holds every result to explicit baselines and negative controls.

EDUCATION

Ph.D., Electrical & Computer Engineering, Georgia Tech, GPA 3.8/4.0 — Aug 2020 – Aug 2026
M.S., Electrical & Computer Engineering, Georgia Tech, GPA 3.9/4.0 — Aug 2023
B.S., Electrical Engineering, Ferdowsi University of Mashhad, Iran, GPA 4.0/4.0 — Sep 2019

RESEARCH EXPERIENCE

Researcher, Georgia Tech School of ECE and TReNDS Center, Atlanta, GA — Aug 2020 – Present
Graduate Research Assistant through Aug 2026, continuing research affiliation since the Ph.D.; TReNDS is a tri-university neuroimaging and data-science center.

RnR: causal discovery from noisy, slowly sampled time series (NeurIPS 2026)

- Built RnR, a solver that takes the output of standard causal-discovery methods and corrects the errors slow sampling introduces (indirect influences that appear direct); raised edge F1 by 45% over four widely used methods.
- Weights each input edge by reliability, so contradictory inputs yield the closest consistent causal graph instead of a failure; solved exactly, with a staged search that keeps runtime practical.
- Applied it to brain-imaging time series from 311 subjects (patients vs. controls) with collaborators at Yale and the University of Virginia: found more directed group-difference connections than five standard methods, and tighter preprocessing raised cross-subject reproducibility (Jaccard overlap) from 0.31 to 0.46.
- Audited the solver pipeline end to end, found and fixed silent correctness bugs, and added unit tests and a regression checklist that every new analysis pipeline must pass.

Transformers for causal discovery from slowly sampled time series (paper in preparation)

- Trained a transformer on ~1M simulated multichannel recordings to score directed links directly from slowly sampled data (AUROC 0.71–0.77 across sampling rates); trained on A100/H100 GPUs via SLURM.
- Built an evaluation harness that requires every model to beat a no-data baseline and shuffled-time controls; it caught a data-leakage bug in a synthetic benchmark before any result was reported.

LLM-guided selection among equivalent causal graphs (ongoing)

- Fine-tuned open-weight LLMs (Mistral-7B, Llama-3-8B; LoRA/QLoRA with Hugging Face PEFT) to choose the true causal graph among candidates the data alone cannot distinguish, using chain-of-thought reasoning and domain knowledge; built the simulator-backed training set and the benchmark (selection accuracy, edge F1).

Scalable causal discovery and multi-source integration

- Scaled causal graph learning from the 6-node ceiling of prior exact methods to graphs with 100+ nodes by decomposing each graph into its strongly connected components, so cost grows linearly with their number.
- Co-developed ION-C, which merges datasets that each measure a different subset of variables into one consistent causal graph; proved correctness and completeness and scaled from 4–6 to 25+ nodes.

Exact causal discovery under an unknown sampling rate (GRACE-C, ICLR 2023; CLeaR 2023)

- Reformulated the problem as a constraint-satisfaction program for an off-the-shelf solver, cutting runtime from 17 hours to 6 seconds; proved correctness and completeness and matched the provably correct baseline on 1,000 test graphs.
- Co-developed theory showing that the choice of sampling interval, including mixing two sampling rates, can shrink the set of causal models consistent with the data by up to 10,000×: how you sample determines what you can identify.
- Released the methods in the lab's open-source gunfolds package (PyPI) with CI/CD and documentation.

PUBLICATIONS

- **M. Abavisani** et al. RnR: A Meta-Solver for Causal Discovery in Undersampled Time Series. *NeurIPS 2026*.
- **M. Abavisani**, K. Solovyeva, D. Danks, G. Pearlson, V. Calhoun, S. Plis. Recovering Directional Brain Networks under Temporal Undersampling: An Application to Schizophrenia. Under review, *Network Neuroscience* (bioRxiv 2026.08.09.743708).
- P. Nair, P. Bhandari, **M. Abavisani**, S. Plis, D. Danks. ION-C: Integration of Overlapping Networks via Constraints. Under review, *ICLR 2027*.
- **M. Abavisani**, D. Danks, S. Plis. GRACE-C: Generalized Rate Agnostic Causal Estimation via Constraints. *ICLR 2023* (notable top 25%; oral presentation).
- K. Solovyeva, D. Danks, **M. Abavisani**, S. Plis. Causal Learning through Deliberate Undersampling. *CLeaR 2023* (PMLR).
- A. Naghizadeh, **M. Abavisani**, D. Metaxas. Greedy AutoAugment. *Pattern Recognition Letters*, 2020.

TECHNICAL SKILLS

Languages: Python (primary), C++, Java, R, SQL, Bash

ML & statistics: PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, pandas, statsmodels; hypothesis testing, bootstrap, false discovery rate control

Causal inference & time series: causal discovery, Granger causality, VAR, PCMCI, multivariate time-series modeling, evaluation design with pre-specified baselines and negative controls

LLM systems: Hugging Face Transformers and PEFT (LoRA/QLoRA), retrieval-augmented generation, multi-agent pipelines, evaluation harnesses, local inference (Ollama), FastAPI

Graphs & optimization: graph algorithms, combinatorial optimization (answer-set programming, clingo)

Infrastructure: Linux, SLURM HPC clusters (A100/H100 GPUs), Docker, Git, GitHub Actions CI/CD, pytest, AI-assisted development (Claude Code)

SELECTED PROJECTS

News-screening pipeline on local LLMs Jul 2026 – Present

- Seven-stage LLM pipeline that screens market news for a stock portfolio and has run unattended every trading day since Jul 2026: two local open-weight models (Qwen3.5-122B, Gemma 4 on Ollama) classify each item independently and only agreeing calls pass; checkpointed so one bad input never stops a run.
- Multi-model ensemble served through FastAPI as an OpenAI-compatible API: several models answer, rank each other's answers blind, and a final model writes the synthesis.

Local LLM Monitor Jul 2026

- macOS dashboard for local LLM serving (CPU and memory pressure, latency, token counts via a metadata-only proxy); every metric is labeled Measured, Reported, Inferred or Unavailable so missing data never reads as zero.

HONORS

- Ranked 537th of 1,000,000+ (top 0.06%) in Iran's national university entrance exam; ranked 3rd in ACM programming competitions